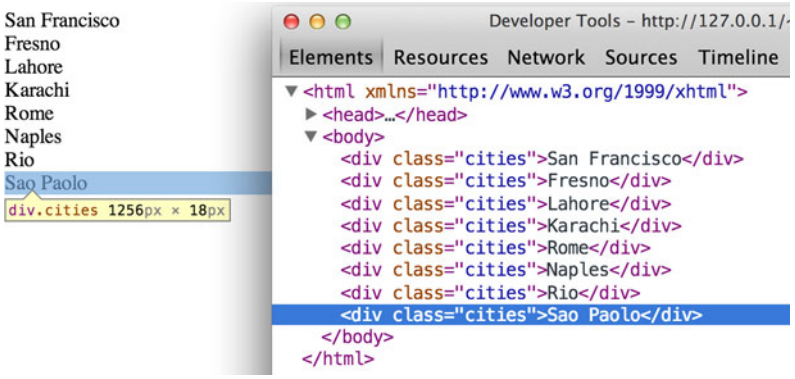
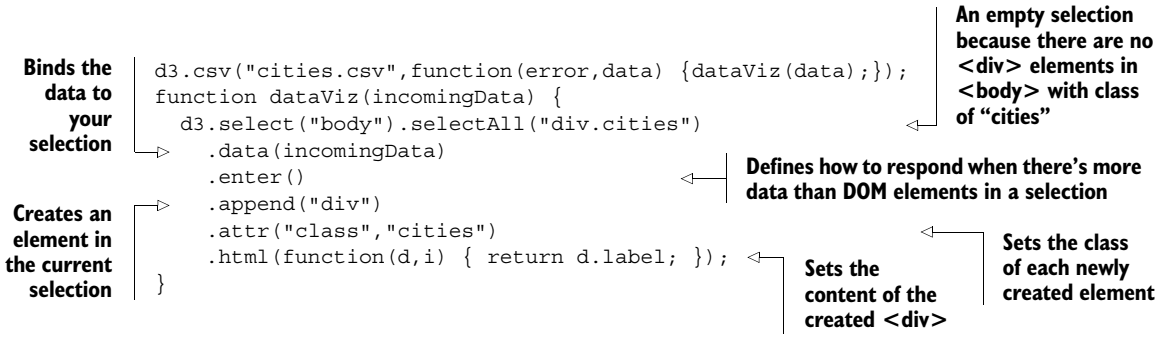


Figure 2.11 When our selection binds the cities.csv data to our web page, it creates eight new divs, each of which is classed with `"cities"` and with content drawn from our data.



The selection and binding procedure shown here is a common pattern throughout the rest of this book. A subselection is created when you first select one element and then select the elements underneath it, which you'll see in more detail later. First, let's take a look at each individual part of this example.

D3.SELECTALL()

The first part of any selection is `d3.select()` or `d3.selectAll()` with a CSS identifier that corresponds to a part of the DOM. Often no elements match the identifier, which is referred to as an *empty selection*, because you want to create new elements on the page using the `.enter()` function. You can make a selection on a selection to designate how to create and modify child elements of a specific DOM element. Note that a subselection won't automatically generate a parent. The parent must already exist, or you'll need to create one using `.append()`.

.DATA()

Here you associate an array with the DOM elements you selected. Each city in our dataset is associated with a DOM element in the selection, and that associated data is stored